Incident report: false declines after the fraud-scorer v3.4 release

Document: INC-201

Date: 2026-01-22

Services: fraud-scorer, payments-api

Summary

On 2026-01-22, fraud-scorer model v3.4 was deployed at 10:15 UTC as part of CHG-1011. The new model exhibited significantly elevated decline rates on legitimate card payments, triggering a SEV2 incident. The model was rolled back at 11:40 UTC to v3.3. Incident commander Lucas Moreau led the response.

Impact

The elevated decline rate increased from 2.1% to 9.8% during the incident window. Legitimate merchants experienced payment failures, reducing transaction success rates and creating customer friction. The fraud-scorer service remained operational but was miscalibrated, causing unnecessary payment rejections.

Timeline

- 10:15 UTC: fraud-scorer model v3.4 deployed
- 10:28 UTC: monitoring alerts triggered for elevated decline rates
- 10:35 UTC: incident declared; STS-101 status page updated
- 10:42 UTC: root cause identified as v3.4 model behavior
- 11:40 UTC: rolled back at 11:40 UTC to v3.3 using RB-06
- 11:52 UTC: decline rates returned to baseline; incident resolved

Root Cause

fraud-scorer model v3.4 applied overly aggressive scoring thresholds that rejected legitimate payment patterns. The model-level analysis to determine why v3.4 declined legitimate payments remains an open follow-up item.

Resolution

The incident was resolved by rolling back fraud-scorer to model v3.3 following RB-06. Decline rates normalized within minutes of rollback completion. No data loss or ledger inconsistencies occurred.

Follow-up Actions

- Complete model-level analysis of v3.4 behavior before any re-release attempt
- Review CHG-1011 change controls and pre-deployment validation procedures
- Implement canary deployment for future fraud-scorer model releases to catch calibration issues earlier
- Establish tighter alerting thresholds for decline rate anomalies